Exercise 5: Rule-Based Learning (Practice)

Given training dataset

Example	Weekday	Mother in car	Mood	Time	Class
1	Monday	no	easygoing	evening	Yes
2	Monday	no	annoyed	evening	No
3	Saturday	yes	easygoing	lunchtime	No
4	Monday	no	easygoing	morning	Yes

(a) Apply the Find-S algorithm

Find-S finds the most specific hypothesis that covers all positive training examples. It ignores negative examples.

Step 1 – Initialize:

$$S_0 = (\varnothing, \varnothing, \varnothing, \varnothing)$$

Step 2 – Example 1 (Positive): (Monday, no, easygoing, evening). Set the hypothesis to this first positive example.

$$S_1 = (\text{Monday, no, easygoing, evening})$$

Step 3 – Example 2 (Negative): (Monday, no, annoyed, evening). Ignore the negative example; S does not change.

$$S_2 = (\text{Monday, no, easygoing, evening})$$

Step 4 – Example 3 (Negative): (Saturday, yes, easygoing, lunchtime). Ignore the negative example; S does not change.

$$S_3 = (\text{Monday, no, easygoing, evening})$$

Step 5 – Example 4 (Positive): (Monday, no, easygoing, morning). The first three attributes match, but the time differs (evening versus morning). Replace the time restriction with ?.

$$S_4 = (\text{Monday, no, easygoing, ?})$$

Final answer (a): S = (Monday, no, easygoing, ?)

(b) Apply the Candidate-Elimination algorithm

Candidate-Elimination maintains two boundaries: S, the most specific boundary, and G, the most general boundary. A positive example generalizes S and removes inconsistent hypotheses from G. A negative example specializes G and removes inconsistent hypotheses from S.

Initialization

$$S_0 = (\varnothing, \varnothing, \varnothing, \varnothing) \qquad G_0 = \{(?, ?, ?, ?)\}$$

Step 1 – Example 1 (Positive): (Monday, no, easygoing, evening). Generalize S to cover the positive example. G remains unchanged.

$$S_1 = (\text{Monday, no, easygoing, evening})$$
$$G_1 = \{(?, ?, ?, ?)\}$$

Step 2 – Example 2 (Negative): (Monday, no, annoyed, evening). Specialize G minimally so it excludes this example while remaining consistent with S. Only the mood differs from the positive example, so require Mood = easygoing.

$$S_2 = (\text{Monday, no, easygoing, evening})$$
$$G_2 = \{(?, ?, \text{easygoing}, ?)\}$$

Step 3 – Example 3 (Negative): (Saturday, yes, easygoing, lunchtime). This example is covered by the current G hypothesis. To exclude it while retaining the positive example, create the minimal specializations Weekday = Monday, Mother = no, or Time = evening.

$$S_3 = (\text{Monday, no, easygoing, evening})$$

$$G_3 = \{ (\text{Monday, ?, easygoing, ?}),\ \ (\text{?, no, easygoing, ?}),\ \ (\text{?, ?, easygoing, evening}) \}$$

Step 4 – Example 4 (Positive): (Monday, no, easygoing, morning). Generalize S by replacing the differing time value with ?. Remove from G any hypothesis that does not cover this positive example. The hypothesis requiring evening is removed.

$$S_4 = (\text{Monday, no, easygoing, ?})$$

$$G_4 = \{ (\text{Monday, ?, easygoing, ?}),\ \ (\text{?, no, easygoing, ?}) \}$$

Final answer (b)

Specific boundary H_S:

$$H_S = \{ (\text{Monday, no, easygoing, ?}) \}$$

General boundary H_G:

$$H_G = \{ (\text{Monday, ?, easygoing, ?}),\ \ (\text{?, no, easygoing, ?}) \}$$

(c) Determine the version space H_D

The version space contains every hypothesis that is at least as general as the specific boundary and at least as specific as one of the general-boundary hypotheses. In other words, these are all hypotheses consistent with all four training examples.

The three hypotheses are:

1. (Monday, no, easygoing, ?)
2. (Monday, ?, easygoing, ?)
3. (?, no, easygoing, ?)

Final answer (c):

$$H_D = \{ (\text{Monday, no, easygoing, ?}), (\text{Monday, ?, easygoing, ?}), (\text{?, no, easygoing, ?}) \}$$

Summary of final results

Part	Result
(a) Find-S	S = (Monday, no, easygoing, ?)
(b) Specific boundary	H_S = {(Monday, no, easygoing, ?)}
(b) General boundary	H_G = {(Monday, ?, easygoing, ?), (?, no, easygoing, ?)}
(c) Version space	H_D contains the three hypotheses listed above.